

Fourier Analysis

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This document contains unofficial student-made notes for the course Introduction to Fourier Analysis taught by Gianluca Giacchi in Winter 2025/2026 at the Università della Svizzera italiana. If you spot an error, please report it to fabianlucasbosshard@gmail.com. The up-to-date PDF is available at <https://fabianbosshard.github.io/usi-informatics-course-summaries/>. The L^AT_EX source is available at <https://github.com/fabianbosshard/usi-informatics-course-summaries>.

1 Preliminaries

Greek letters

Name	Low.	Var.	Upp.
alpha	α		A
beta	β		B
gamma	γ		Γ
delta	δ		Δ
epsilon	ϵ	ε	E
zeta	ζ		Z
eta	η		H
theta	θ	ϑ	Θ
iota	ι		I
kappa	κ	$\varkappa$	K
lambda	λ		Λ
mu	μ		M
nu	ν		N
xi	ξ		Ξ
omicron	o		O
pi	π	ϖ	Π
rho	ρ	ϱ	P
sigma	σ	ς	Σ
tau	τ		T
upsilon	υ		Υ
phi	ϕ	φ	Φ
chi	χ		X
psi	ψ		Ψ
omega	ω		Ω

Algebraic properties

- Associative: $(a \circ b) \circ c = a \circ (b \circ c)$ *parentheses (grouping) do not matter*
- Commutative: $a \circ b = b \circ a$ *order does not matter*
- Distributive: $a \cdot (b + c) = a \cdot b + a \cdot c$
 $(a + b) \cdot c = ac + bc$

Trigonometric identities

- $\cos(-y) = \cos(y)$
- $\sin(-y) = -\sin(y)$
- $\cos(a + b) = \cos(a)\cos(b) - \sin(a)\sin(b)$
- $\sin(a + b) = \sin(a)\cos(b) + \cos(a)\sin(b)$

1.1 Complex functions

If $f : \mathbb{R} \rightarrow \mathbb{C}$, we may write $f(x) = \Re(f(x)) + \imath \Im(f(x))$. We have $|f(x)|^2 = f(x)\overline{f(x)}$, where $\overline{f(x)} = \Re(f(x)) - \imath \Im(f(x))$.

The imaginary part of $\log z := \log |z| + \imath \arg(z)$ is not unique (see Figure 1).

Consider $\mathbb{C}^{-*} := \mathbb{C} \setminus \{z \in \mathbb{R} \mid z \leq 0\}$. To obtain uniqueness, we define

$$\begin{aligned} \text{Log} : \mathbb{C}^{-*} &\rightarrow \mathbb{C} \\ z &\mapsto \log |z| + \imath \text{Arg}(z) \end{aligned}$$

where $\text{Arg}(z) := \arg(z) \cap (-\pi, \pi)$. This is called the *principal branch* of the logarithm.

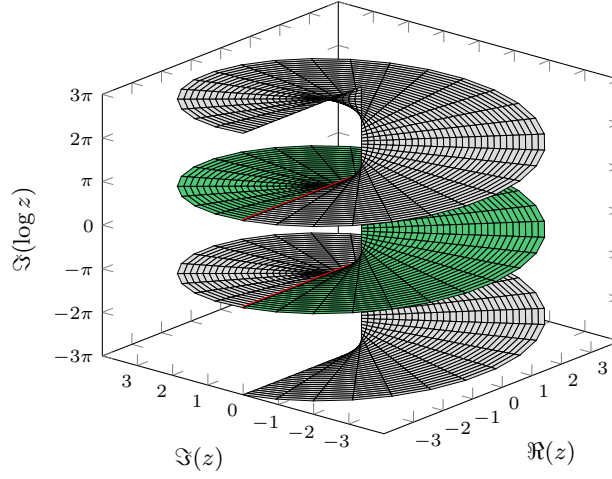


Figure 1: Imaginary part of $\log(z)$ as a function of $\Re(z)$ and $\Im(z)$. Green corresponds to the principal branch, red lines correspond to the branch cut along the negative real axis.

1.2 Lebesgue integration

$\mathcal{R}([a, b])$ denotes the class of Riemann integrable functions on $[a, b]$.

Theorem 1.1. Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded.

- If $f \in \mathcal{R}([a, b])$, then the Lebesgue integral of $|f|$ is finite. Moreover, the Riemann integral of f equals the Lebesgue integral of f .
- $f \in \mathcal{R}([a, b])$ if and only if f is continuous except on a set of measure zero. $\triangleleft$

The motivation for replacing Riemann with Lebesgue theory is that Lebesgue integrals are particularly well-suited for interchanging limits and integration, the order of integration and derivatives and integration.

Theorem 1.2 (Dominated convergence). Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of functions in $L^1(\mathbb{R})$ such that

- there exists $f \in L^1(\mathbb{R})$ so that $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ for almost all $x \in \mathbb{R}$
- there exists $g \in L^1(\mathbb{R})$ such that $|f_n(x)| \leq |g(x)|$ for almost all $x \in \mathbb{R}$ and all $n \in \mathbb{N}$

Then

$$\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} f_n(x) dx = \int_{-\infty}^{\infty} f(x) dx \quad \triangleleft$$

As mentioned before, we are also free to choose the order of integration. According to Fubini-Tonelli, if $f \in L^1(\mathbb{R}^2)$, we may calculate $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy$ by first integrating with respect to x and then with respect to y or the other way around.

Finally, differentiation and integration can also be interchanged under certain conditions. Indeed,

$$\frac{d}{dt} \int_{-\infty}^{\infty} f(x, t) dx = \int_{-\infty}^{\infty} \frac{\partial}{\partial t} f(x, t) dx \quad (1.1)$$

if certain conditions are met. Generally speaking, if $f(\cdot, t) \in L^1(\mathbb{R})$ for all t , $\frac{\partial f}{\partial t}$ exists and

$$\left| \frac{\partial}{\partial t} f(x, t) \right| \leq |g(x)|$$

for some $g \in L^1(\mathbb{R})$ and every $(x, t) \in \mathbb{R}^2$ outside a set of measure zero, then (1.1) holds.

1.3 Function spaces: C^k , L^p , C_b , C_c^∞

Classically, functions are classified in terms of their regularity. We write $C(\mathbb{R})$ for the space of continuous functions from $\mathbb{R}$ to $\mathbb{C}$. For $k \in \mathbb{N}$, we write $C^k(\mathbb{R})$ for the space of functions from $\mathbb{R}$ to $\mathbb{C}$ whose first k derivatives exist and are continuous.

Example 1.1. The function $f(x) = x^k \operatorname{sgn}(x)$ is in $C^{k-1}(\mathbb{R})$ but not in $C^k(\mathbb{R})$. ◀

The revolution in modern analysis is that functions are classified in terms of their integrability. For $1 \leq p < \infty$, we write $f \in L^p(\mathbb{R})$ if

$$\|f\|_p := \left(\int_{-\infty}^{\infty} |f(x)|^p dx \right)^{1/p} \quad (1.2)$$

is finite. To be precise, we also need the requirement that $f : \mathbb{R} \rightarrow \mathbb{C}$ is measurable.

Remark 1.2. There is also a definition of $L^\infty(\mathbb{R})$, where $f \in L^\infty(\mathbb{R})$ if

$$\|f\|_\infty := \operatorname{ess\,sup}_{x \in \mathbb{R}} |f(x)| \quad (1.3)$$

is finite. The essential supremum is basically a supremum outside a set of measure zero. If f is in $C(\mathbb{R})$, then $\|f\|_\infty = \sup_{x \in \mathbb{R}} |f(x)|$. If, in addition, $f(x) \rightarrow 0$ as $|x| \rightarrow \infty$, i.e. $f \in C_0(\mathbb{R})$, then $\|f\|_\infty = \max_{x \in \mathbb{R}} |f(x)|$. ◀

Remark 1.3. Strictly speaking, $\|\cdot\|_p$ for $1 \leq p \leq \infty$, i.e. eqs. (1.2) and (1.3), is not a norm on functions defined pointwise, but on equivalence classes of functions that are equal almost everywhere. ◀

We write $f \in C_b(\mathbb{R})$ if $f \in C(\mathbb{R})$ and there exists $M > 0$ such that $|f(x)| \leq M$ for all $x \in \mathbb{R}$ (i.e., f is bounded). Observe that $C_b(\mathbb{R}) = L^\infty(\mathbb{R}) \cap C(\mathbb{R})$.

Lebesgue L^p are not closed under product, i.e., $f, g \in L^p(\mathbb{R}) \not\Rightarrow fg \in L^p(\mathbb{R})$. However, if f and g are in conjugate spaces, something can be said about the product fg .

For $1 \leq p \leq \infty$, the *Lebesgue conjugate exponent* of p is the number p' defined by

$$\frac{1}{p} + \frac{1}{p'} = 1$$

where we set $\frac{1}{\infty} := 0$ and $\frac{1}{0} := \infty$.

Theorem 1.3 (Hölder's inequality). If $f \in L^p(\mathbb{R})$ and $g \in L^{p'}(\mathbb{R})$, then $fg \in L^1(\mathbb{R})$ and

$$\|fg\|_1 \leq \|f\|_p \|g\|_{p'} \quad (1.4)$$

◀

Definition 1.1. Let $(X, \|\cdot\|_X)$ be a normed space and $Y \subseteq X$ a subspace of X . We say that Y is *dense* in X if for every $x \in X$ and every $\varepsilon > 0$ there exists $y \in Y$ such that $\|x - y\|_X < \varepsilon$. ◀

Example 1.4. $\mathbb{Q}$ is dense in $\mathbb{R}$. ◀

Example 1.5. Define $C_c^\infty(\mathbb{R}) := \{f \in C^\infty(\mathbb{R}) : \exists R > 0 \text{ such that } f(x) = 0 \text{ for } |x| > R\}$ the space of *compactly supported* smooth functions. $C_c^\infty(\mathbb{R})$ is dense in $L^p(\mathbb{R})$ for every $1 \leq p < \infty$. ◀

Often easier to prove a statement in e.g. $C_c^\infty(\mathbb{R})$ and then extend it to e.g. $L^1(\mathbb{R})$ by invoking e.g. Theorem 1.2.

1.4 Operator theory

Let X, Y be vector spaces (over $\mathbb{C}$). A linear operator $T: X \rightarrow Y$ is a mapping that satisfies

- (i) $T(x_1 + x_2) = T(x_1) + T(x_2)$ for all $x_1, x_2 \in X$ (additivity)
- (ii) $T(\lambda x) = \lambda T(x)$ for all $x \in X$ and $\lambda \in \mathbb{C}$ (homogeneity)

We often write Tx instead of $T(x)$, if it does not cause ambiguity.

Example 1.6. If $X = \mathbb{C}^n$ and $Y = \mathbb{C}^m$, then T is linear iff there exists a matrix $\underline{M} \in \mathbb{C}^{m \times n}$ such that $T(\mathbf{x}) = \underline{M}\mathbf{x}$ for all $\mathbf{x} \in \mathbb{C}^n$. ◀

Linear operators between finite-dimensional spaces are always identified with their matrix. They are fundamentally trivial. For instance, they are always continuous, a property not shared by linear operators between infinite-dimensional spaces.

Definition 1.2. Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces. A linear operator $T: X \rightarrow Y$ is *bounded* if there exists $C > 0$ such that

$$\|Tx\|_Y \leq C\|x\|_X \quad (1.5)$$

for all $x \in X$. The infimum of the constants C for which (1.5) holds is called the *operator norm* of T . ◀

Roughly speaking, T is bounded when bounded sets in X cannot grow too much under the action of T . Observe that if $(x_n)_{n \in \mathbb{N}}$ is a sequence in X converging to x , then

$$\|Tx_n - Tx\|_Y = \|T(x_n - x)\|_Y \leq C\|x_n - x\|_X \rightarrow 0$$

as $n \rightarrow \infty$, i.e. $\lim_{n \rightarrow \infty} Tx_n = Tx$ in Y , implying that T is continuous. This is a characterization of bounded linear operators between normed spaces.

A normed space $(X, \|\cdot\|_X)$ is *complete* if whenever $(x_n)_{n \in \mathbb{N}}$ satisfies

$$\lim_{n, m \rightarrow \infty} \|x_n - x_m\|_X = 0$$

there exists $x \in X$ such that $\lim_{n \rightarrow \infty} \|x - x_n\|_X = 0$, i.e. $x = \lim_{n \rightarrow \infty} x_n$ in X . A complete normed vector space is called a *Banach space*. Loosely speaking, X is complete if it has no holes.

Example 1.7. $\mathbb{Q}$ is not complete, $\mathbb{R}$ and $\mathbb{C}$ are complete. ◀

Example 1.8. $(L^p(\mathbb{R}), \|\cdot\|_p)$ is complete for every $1 \leq p \leq \infty$. ◀

Remark 1.9. Finite-dimensional vector spaces (over a complete field) are always complete, with respect to any norm defined on them. ◀

Theorem 1.4. Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces, with Y complete. Let Z be a dense subspace of X and $T: Z \rightarrow Y$ be a linear bounded operator. Then, there exists a unique linear bounded operator $\tilde{T}: X \rightarrow Y$ such that $\tilde{T}x = Tx$ for every $x \in Z$. When there is no risk of ambiguity, we often write T instead of $\tilde{T}$ for this *extended operator*. ◁

2 Fourier Series

We write $f \in L^2([-\pi, \pi])$ if $f : \mathbb{R} \rightarrow \mathbb{C}$ is 2π -periodic and

$$\|f\|_2 = \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x)|^2 dx \right)^{\frac{1}{2}} \quad (2.1)$$

is finite, i.e. $\|f\|_2 < \infty$.

Functions in $L^2([-\pi, \pi])$ are called finite-energy signals, the norm $\|f\|_2$ is the energy of f . Observe that we normalized the integral in (2.1) by means of the factor $\frac{1}{2\pi}$. Loosely speaking, we change the unit of measurement so that $\text{length}([-\pi, \pi]) = \frac{1}{2\pi} \int_{-\pi}^{\pi} 1 dx = 1$.

Different from other L^p spaces, $L^2([-\pi, \pi])$ has additional structure provided by the (sesquilinear) inner product

$$\langle f, g \rangle_{L^2} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \overline{g(x)} dx \quad (2.2)$$

for $f, g \in L^2([-\pi, \pi])$. (2.2) induces (2.1) since $\|f\|_2 = \sqrt{\langle f, f \rangle_{L^2}}$.

$L^2([-\pi, \pi])$ is not finite-dimensional, i.e. it does not admit a finite basis. However, it does have a countable orthonormal (Schauder) basis, i.e. there exists an infinite sequence of functions $(e_n)_{n \in \mathbb{Z}} \subseteq L^2([-\pi, \pi])$ such that $\langle e_n, e_m \rangle_{L^2} = \delta_{n,m}$ and for every $f \in L^2([-\pi, \pi])$ there exist unique $a_n \in \mathbb{C}$ such that

$$f \stackrel{L^2}{=} \sum_n a_n e_n \quad (2.3)$$

where the symbol $\stackrel{L^2}{=}$ means that

$$\lim_{N \rightarrow \infty} \left\| f - \sum_{n=-N}^N a_n e_n \right\|_2 = 0 \quad (2.4)$$

i.e. the series converges in the L^2 norm. We can compute the coefficients explicitly since

$$\langle f, e_m \rangle_{L^2} = \left\langle \sum_n a_n e_n, e_m \right\rangle_{L^2} = \sum_n a_n \langle e_n, e_m \rangle_{L^2} = a_m$$

where we used the component-wise continuity of $\langle \cdot, \cdot \rangle_{L^2}$ to intertwine series and inner products.

The prototype of such an orthonormal basis are the functions $e_n(x) = e^{inx}$ with $n \in \mathbb{Z}$: For every $f \in L^2([-\pi, \pi])$,

$$f(x) \stackrel{L^2}{=} \sum_{n=-\infty}^{\infty} \hat{f}(n) e^{inx} \quad (2.5)$$

where, for $n \in \mathbb{Z}$,

$$\hat{f}(n) = \langle f, e_n \rangle_{L^2} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \quad (2.6)$$

is the n -th Fourier coefficient of f . (2.5) is the Fourier series of f .

Theorem 2.1. For every $f \in L^2([-\pi, \pi])$, we have

$$\|f\|_2^2 = \sum_{n=-\infty}^{\infty} |\hat{f}(n)|^2 \quad (2.7)$$

which is known as **Parseval's theorem**. ◀

Proof. By the properties of the inner product,

$$\begin{aligned} \|f\|_2^2 &= \langle f, f \rangle_{L^2} = \left\langle \sum_n \hat{f}(n) e_n, \sum_m \hat{f}(m) e_m \right\rangle_{L^2} \\ &= \sum_n \sum_m \hat{f}(n) \overline{\hat{f}(m)} \underbrace{\langle e_n, e_m \rangle_{L^2}}_{\delta_{n,m}} = \sum_n |\hat{f}(n)|^2 \end{aligned}$$

where we again used the component-wise continuity of $\langle \cdot, \cdot \rangle_{L^2}$. ◻

Corollary 2.2. If $f \in L^2([-\pi, \pi])$ has $\hat{f}(n) = 0$ for all $n \in \mathbb{Z}$, then $f = 0$ in $L^2([-\pi, \pi])$. $\triangleleft$

Proof. $\hat{f}(n) = 0 \quad \forall n \in \mathbb{Z} \quad \xrightarrow{(2.7)} \quad \|f\|_2^2 = 0 \quad \Rightarrow \quad f = 0_{L^2([-\pi, \pi])}$ $\square$

Example 2.1. The C^∞ function e^{-1/x^2} is a prototypical example of a non-zero function with all vanishing Taylor coefficients at 0. Corollary 2.2 shows that these pathologies do not occur for Fourier series. $\blacktriangleleft$

Corollary 2.3 (Riemann-Lebesgue lemma). We have

$$\lim_{|n| \rightarrow \infty} \hat{f}(n) = 0 \quad (2.8)$$

for every $f \in L^2([-\pi, \pi])$. $\triangleleft$

Proof. $f \in L^2([-\pi, \pi]) \Rightarrow \|f\|_2^2 \stackrel{(2.7)}{=} \sum_n |\hat{f}(n)|^2 < \infty \Rightarrow \lim_{|n| \rightarrow \infty} |\hat{f}(n)|^2 = 0$ $\square$

Lebesgue spaces on finite measure spaces satisfy the inclusion relations

$$L^q \subseteq L^p \quad (2.9)$$

for $1 \leq p < q \leq \infty$. Thus, $L^1([-\pi, \pi])$ is the largest of the L^p spaces on $[-\pi, \pi]$. Furthermore, the Fourier coefficients are well-defined for functions in $L^1([-\pi, \pi])$, since

$$|\hat{f}(n)| = \left| \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \right| \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x)| dx = \|f\|_1$$

exists and is finite for every $f \in L^1([-\pi, \pi])$ and $n \in \mathbb{Z}$. Proving results such as Corollaries 2.2 and 2.3 is considerably simplified in the L^2 setting thanks to the inner product. Nevertheless, together with most of the results from this section, they also hold for $L^1([-\pi, \pi])$. In fact, except for (2.3) and (2.7), all statements from this section are already true in $L^1([-\pi, \pi])$, but need to be proved by different methods.

3 Fourier Transform on L^1

For $f \in L^1(\mathbb{R})$ the Fourier transform

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i \xi x} dx \quad (3.1)$$

converges for every $\xi \in \mathbb{R}$. Moreover,

$$|\hat{f}(\xi)| = \left| \int_{-\infty}^{\infty} f(x) e^{-2\pi i \xi x} dx \right| \leq \int_{-\infty}^{\infty} |f(x)| \underbrace{|e^{-2\pi i \xi x}|}_{=1} dx = \|f\|_1 \quad (3.2)$$

for any $\xi \in \mathbb{R}$, implying that $\hat{f}$ is in $L^\infty(\mathbb{R})$ and the operator

$$\mathcal{F} : f \in L^1(\mathbb{R}) \mapsto \hat{f} \in L^\infty(\mathbb{R})$$

is bounded with

$$\|\hat{f}\|_\infty \leq \|f\|_1 \quad (3.3)$$

Definition 3.1. The operator $\mathcal{F}$ defined above is called Fourier transform, and the function $\hat{f}$ is the Fourier transform of f . ◀

Definition 3.2 (Pointwise continuity). Let $f : \mathbb{R} \rightarrow \mathbb{C}$ and $x_0 \in \mathbb{R}$. We say f is continuous at x_0 if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon \quad \blacktriangleleft$$

Definition 3.3 (Uniform continuity). Let $f : \mathbb{R} \rightarrow \mathbb{C}$. We say f is uniformly continuous on $\mathbb{R}$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, y \in \mathbb{R}$

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon \quad \blacktriangleleft$$

Remark 3.1. Uniform continuity is stronger than pointwise continuity and implies it. ◀

Fact 3.1. The Fourier transform is a bounded operator from $L^1(\mathbb{R})$ to $L^\infty(\mathbb{R})$. Moreover, $\hat{f}$ is a uniformly continuous function on $\mathbb{R}$ for every $f \in L^1(\mathbb{R})$. ◁

Proof. Boundedness follows from (3.2). For uniform continuity, consider $|\hat{f}(\xi + h) - \hat{f}(\xi)|$, apply Theorem 1.2 and observe that the obtained expression goes to zero as $h \rightarrow 0$ and does not depend on ξ . ◻

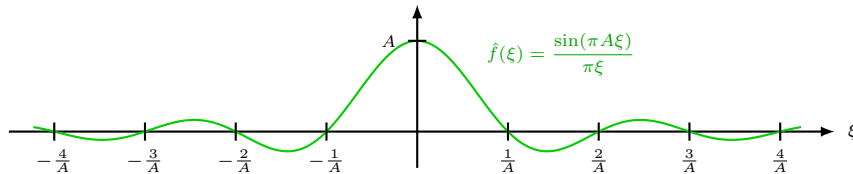
Example 3.2 (Characteristic function). Let

$$f(x) = \chi_{[-A/2, A/2]}(x) := \begin{cases} 1, & x \in [-A/2, A/2] \\ 0, & \text{otherwise} \end{cases} \quad (3.4)$$

be the characteristic function of the interval $[-A/2, A/2]$, with $A > 0$. Then for every $\xi \neq 0$,

$$\hat{f}(\xi) = \int_{-A/2}^{A/2} e^{-2\pi i \xi x} dx = \frac{\sin(\pi A \xi)}{\pi \xi}$$

and $\hat{f}(0) = \int_{-A/2}^{A/2} 1 dx = A$.



Thus we conclude that $\hat{f}(\xi) = \frac{\sin(\pi A \xi)}{\pi \xi}$, where we implicitly consider its continuous continuation. In particular, for $A = 1$, we have

$$\hat{f}(\xi) = \frac{\sin(\pi \xi)}{\pi \xi} =: \text{sinc}(\xi) \quad (3.5)$$

which is called *cardinal sine* or *sinc function*. ◀

Example 3.3 (Gaussian). The Fourier transform of $f(x) = e^{-\pi x^2}$ is $\hat{f}(\xi) = e^{-\pi \xi^2}$. ◀

Example 3.4. The Fourier transform of $f(x) = e^{-2|x|}$ is $\hat{f}(\xi) = \frac{1}{1 + \pi^2 \xi^2}$. ◀

3.1 Important operators in harmonic analysis

Definition 3.4.

$$\text{translation:} \quad T_{x_0}f(x) = f(x - x_0), \quad x_0 \in \mathbb{R} \quad (3.6)$$

$$\text{modulation:} \quad M_{\xi_0}f(x) = e^{2\pi i \xi_0 x} f(x), \quad \xi_0 \in \mathbb{R} \quad (3.7)$$

$$\text{upper dilation:} \quad f^\lambda(x) = f(\lambda x), \quad \lambda > 0 \quad (3.8)$$

$$\text{lower dilation:} \quad f_\lambda(x) = \frac{1}{\lambda} f(x/\lambda), \quad \lambda > 0 \quad (3.9)$$

◀

3.1.1 Duality

Let $f \in L^1(\mathbb{R})$ and $\lambda > 0$.

Translation/Modulation:

$$\widehat{(T_{x_0}f)}(\xi) = M_{-x_0}\hat{f}(\xi) \quad (3.10)$$

$$\widehat{(M_{\xi_0}f)}(\xi) = T_{\xi_0}\hat{f}(\xi) \quad (3.11)$$

Dilation:

$$\widehat{(f^\lambda)}(\xi) = (\hat{f})_\lambda(\xi) \quad (3.12)$$

$$\widehat{(f_\lambda)}(\xi) = (\hat{f})^\lambda(\xi) \quad (3.13)$$

(3.12) and (3.13) are a prelude to the *uncertainty principle*, which states that a function and its Fourier transform cannot both be arbitrarily well localized. Compressing a function in the time domain causes its Fourier transform to expand in the frequency domain and vice versa.

3.2 Convolution

For $f, g: \mathbb{R} \rightarrow \mathbb{C}$, we define the convolution

$$(f * g)(x) := \int_{-\infty}^{\infty} f(y)g(x - y)dy \quad (3.14)$$

If $f, g \in L^1(\mathbb{R})$, then $\|f * g\|_1 \leq \|f\|_1\|g\|_1$, so $f * g \in L^1(\mathbb{R})$. Furthermore, the convolution is commutative, associative and distributive.

Theorem 3.2. If $f \in L^1(\mathbb{R})$, $g \in L^p(\mathbb{R})$ for $1 \leq p \leq \infty$, then

$$\|f * g\|_p \leq \|f\|_1\|g\|_p \quad (3.15)$$

which is known as **Young's inequality**.

◀

Theorem 3.3. The Fourier transform turns convolution into pointwise multiplication

$$\widehat{(f * g)}(\xi) = \hat{f}(\xi) \cdot \hat{g}(\xi) \quad (3.16)$$

for every $f, g \in L^1(\mathbb{R})$ and $\xi \in \mathbb{R}$.

◀

Proof. Plug (3.14) into (3.1), use Fubini, substitute $x - y = z$ and $x = z + y$. ◻

3.3 Differentiation

Define

$$C_0(\mathbb{R}) := \left\{ f \in C(\mathbb{R}) : \lim_{|x| \rightarrow \infty} f(x) = 0 \right\} \quad (3.17)$$

i.e. $f \in C_0(\mathbb{R})$ if f is continuous and decays at infinity.

It is widely known that the normed space $(C_0(\mathbb{R}), \|\cdot\|_\infty)$ is complete, i.e. a Banach space.

Theorem 3.4. Let $f \in L^1(\mathbb{R})$.

(i) If $xf \in L^1(\mathbb{R})$, then $\hat{f} \in C^1(\mathbb{R})$ and

$$\hat{f}'(\xi) = (-2\pi i x f)(\xi) \quad (3.18)$$

(ii) If $f \in C^1(\mathbb{R}) \cap C_0(\mathbb{R})$ and $f' \in L^1(\mathbb{R})$, then

$$\widehat{(f')}(\xi) = 2\pi i \xi \hat{f}(\xi) \quad (3.19)$$

◁

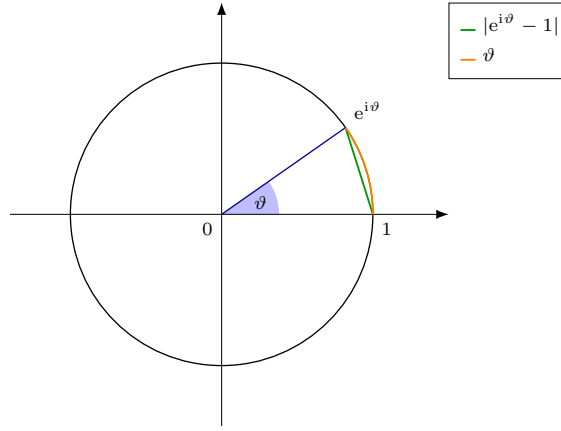


Figure 2: Inequality $|e^{i\vartheta} - 1| \leq \vartheta$ for $\vartheta > 0$.

Proof. (i) Consider $\frac{\hat{f}(\xi+h) - \hat{f}(\xi)}{h}$, factor out the common exponential, so that the dependence on h appears only through $\frac{e^{-2\pi i h x} - 1}{h}$. Using Figure 2 with $\vartheta = -2\pi h x$, this quotient is dominated by a constant multiple of $|xf(x)|$. Since $xf \in L^1(\mathbb{R})$ by assumption, this provides an L^1 -majorant, so we apply Theorem 1.2. This proves that $\hat{f}$ is differentiable and that the derivative is given by (3.18). Moreover, $\hat{f}'$ is a Fourier transform of an L^1 -function, hence is (uniformly) continuous by Fact 3.1, so $\hat{f} \in C^1(\mathbb{R})$.

(ii) Integrate by parts, using the fact that f vanishes at infinity. □

By iterating (3.18), we obtain

Theorem 3.5. Let $f \in L^1(\mathbb{R}) \cap C^k(\mathbb{R})$, $k \geq 1$. If

(i) $f^{(j)} \in L^1(\mathbb{R})$ for every $j \leq k$

(ii) $f^{(j)} \in C_0(\mathbb{R})$ for every $j \leq k - 1$

then

$$\widehat{(f^{(j)})}(\xi) = (2\pi i \xi)^j \hat{f}(\xi) \quad (3.20)$$

for every $j \leq k$ and $\xi \in \mathbb{R}$. ◁

As a corollary, if $f \in C_c^\infty(\mathbb{R})$, then $\hat{f} \in C^\infty(\mathbb{R})$ and (3.20) holds for every $j \in \mathbb{N}$. In particular, if $f \in C_c^\infty(\mathbb{R})$, by choosing $j = 1$, we get

$$|\hat{f}(\xi)| \leq \frac{|\widehat{(f')}(\xi)|}{2\pi|\xi|} \stackrel{(3.3)}{\leq} \frac{\|f'\|_1}{|\xi|} \rightarrow 0 \quad (3.21)$$

as $|\xi| \rightarrow \infty$. If $f \in L^1(\mathbb{R})$, by the density of $C_c^\infty(\mathbb{R})$ in $L^1(\mathbb{R})$, there exists a sequence of functions $(f_i)_{i \in \mathbb{N}}$, $f_i \in C_c^\infty(\mathbb{R})$ such that $\lim_{j \rightarrow \infty} \|f - f_j\|_1 = 0$. Then,

$$\|\hat{f} - \hat{f}_j\|_\infty \stackrel{(3.3)}{\leq} \|f - f_j\|_1 \rightarrow 0$$

as $j \rightarrow \infty$. Thus, $\hat{f} \in C_0(\mathbb{R})$, since every $\hat{f}_j \in C_0(\mathbb{R})$ and $C_0(\mathbb{R})$ is complete with respect to $\|\cdot\|_\infty$. In essence:

Theorem 3.6 (Riemann-Lebesgue lemma). If $f \in L^1(\mathbb{R})$, then

$$\hat{f}(\xi) \in C_0(\mathbb{R}) \quad (3.22)$$

i.e., $\hat{f}$ is continuous and decays at infinity. $\triangleleft$

3.4 Inverse

Lemma 3.7. If $f, g \in L^1(\mathbb{R})$, then $\int_{-\infty}^{\infty} \hat{f}(y)g(y)dy = \int_{-\infty}^{\infty} f(x)\hat{g}(x)dx$. $\triangleleft$

Proof. Plug in (3.1) and use Fubini. $\square$

Lemma 3.8. If $u_n \rightarrow f$ in $L^1(\mathbb{R})$ and $u_n(x) \rightarrow g(x)$ for almost every $x \in \mathbb{R}$, then we have $f = g$ almost everywhere. $\triangleleft$

Remark 3.5. Even if $u_n(x) \rightarrow g(x)$ for all $x \in \mathbb{R}$, we still only get $f = g$ almost everywhere, not necessarily everywhere. $\blacktriangleleft$

Theorem 3.9. Let $f, K \in L^1(\mathbb{R})$ and $\int_{-\infty}^{\infty} K(y)dy = 1$. Then

$$\|f - K_\lambda * f\|_1 \rightarrow 0 \quad (3.23)$$

as $\lambda \rightarrow 0$. $\triangleleft$

Theorem 3.9 states that $K_\lambda \rightarrow \delta$ as $\lambda \rightarrow 0$ in the sense of distributions, where $\delta: C(\mathbb{R}) \rightarrow \mathbb{C}$ is the operator $\delta(g) = g(0)$, $g \in C(\mathbb{R})$.

Theorem 3.10 (Inverse). Let $f \in L^1(\mathbb{R})$. Assume that $\hat{f} \in L^1(\mathbb{R})$. Then,

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i \xi x} d\xi \quad (3.24)$$

for almost every $x \in \mathbb{R}$. If $f \in C(\mathbb{R})$, then (3.24) holds for every $x \in \mathbb{R}$. $\triangleleft$

Proof. Let $\lambda > 0$ and $x \in \mathbb{R}$. Consider

$$\phi(\xi) := e^{2\pi i \xi x} e^{-\pi \lambda^2 \xi^2} = M_x \varphi^\lambda(\xi) \quad (3.25)$$

where $\varphi(\xi) = e^{-\pi \xi^2}$ is the Gaussian. By (3.11) and (3.12),

$$\hat{\phi}(y) = T_x \hat{\varphi}_\lambda(y) = T_x \varphi_\lambda(y) = \frac{1}{\lambda} \varphi\left(\frac{y-x}{\lambda}\right) = \frac{1}{\lambda} e^{-\pi |x-y|^2 / \lambda^2} = \varphi_\lambda(x-y) \quad (3.26)$$

where we used the fact that φ is even and that $\hat{\varphi} = \varphi$ (Example 3.3).

Now on the one hand, by Lemma 3.7,

$$\begin{aligned} \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i \xi x} e^{-\pi \lambda^2 \xi^2} d\xi &\stackrel{(3.25)}{=} \int_{-\infty}^{\infty} \hat{f}(\xi) \phi(\xi) d\xi \stackrel{\text{Lemma 3.7}}{=} \int_{-\infty}^{\infty} f(y) \hat{\phi}(y) dy \\ &\stackrel{(3.26)}{=} \int_{-\infty}^{\infty} f(y) \varphi_\lambda(x-y) dy \stackrel{(3.14)}{=} (f * \varphi_\lambda)(x) \end{aligned}$$

and, since $\int_{-\infty}^{\infty} \varphi(y) dy = 1$, by Theorem 3.9 $f * \varphi_\lambda \rightarrow f$ as $\lambda \rightarrow 0$ in $L^1(\mathbb{R})$.

On the other hand, as $\lambda \rightarrow 0$, for every $x \in \mathbb{R}$,

$$\int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i \xi x} e^{-\pi \lambda^2 \xi^2} d\xi \rightarrow \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i \xi x} d\xi$$

by Theorem 1.2 (with $g(\xi) = \hat{f}(\xi)$), which is in $L^1(\mathbb{R})$ by assumption.

Setting e.g. $u_n = f * \varphi_{1/n}$, by Lemma 3.8, we conclude that

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i \xi x} d\xi \quad (3.27)$$

for almost all $x \in \mathbb{R}$. Note that

$$\int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i \xi x} d\xi = \widehat{\hat{f}}(-x)$$

and since $\hat{f} \in L^1(\mathbb{R})$, by Fact 3.1, the right-hand side of (3.27) defines a continuous function. Therefore, if f is also continuous, (3.27) holds for all $x \in \mathbb{R}$. $\square$

Corollary 3.11. The operator $\mathcal{F}: L^1(\mathbb{R}) \rightarrow L^\infty(\mathbb{R})$ is injective. $\triangleleft$

Proof. A linear operator $T: X \rightarrow Y$ is injective if $\ker(T) = \{0\}$, i.e., $Tx = 0 \implies x = 0$. Let $f \in L^1(\mathbb{R})$ and assume that $\hat{f} = 0$. Then,

$$f(x) \stackrel{L^1}{=} \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i \xi x} d\xi = 0$$

where we used the fact that $\hat{f} = 0 \in L^1(\mathbb{R})$ to apply Theorem 3.10. $\square$

Corollary 3.12. Assume that $f, \hat{f} \in L^1(\mathbb{R})$. Then,

$$(\mathcal{F}^2 f)(x) = f(-x) \quad (3.28)$$

in $L^1(\mathbb{R})$. In particular,

$$(\mathcal{F}^{-1} f)(x) = (\mathcal{F} f)(-x) \quad (3.29)$$

and $(\mathcal{F}^4 f)(x) = f(x)$ in $L^1(\mathbb{R})$. $\triangleleft$

Proof. We have

$$(\mathcal{F}^2 f)(x) = (\mathcal{F}(\hat{f}))(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{-2\pi i \xi x} d\xi = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i \xi (-x)} d\xi \stackrel{L^1}{=} f(-x)$$

where we used Theorem 3.10 in the last step.

Therefore also, $(\mathcal{F}^4 f)(x) = (\mathcal{F}^2(\mathcal{F}^2 f))(x) = (\mathcal{F}^2 f)(-x) = f(x)$ and $\mathcal{F}^{-1} f(x) = \mathcal{F}^{-1}(\mathcal{F}^2 f)(-x) = \mathcal{F}(-x) = \mathcal{F} f(-x)$ in the sense of $L^1(\mathbb{R})$. $\square$

4 Fourier Transform on L^p

4.1 Fourier Transform on L^2

Unlike the function on $[-\pi, \pi]$ considered in section 2, the $L^p(\mathbb{R})$ spaces do not satisfy (2.9). In particular, $L^1(\mathbb{R})$ is not the largest Lebesgue space on $\mathbb{R}$. Consequently, the results developed for $L^1(\mathbb{R})$ do not extend in an automatic way to $L^2(\mathbb{R})$. Note that the functions $\{e^{2\pi i \xi x}\}_{\xi \in \mathbb{R}}$ are not even in $L^2(\mathbb{R})$, whereas $\{e^{inx}\}_{n \in \mathbb{Z}}$ was an orthonormal basis of $L^2([-\pi, \pi])$. Instead of using orthonormal bases, the results for $L^2(\mathbb{R})$ are proven using functional analysis and operator theory, extending them by density from $L^1(\mathbb{R})$.

4.1.1 Inner product and Hilbert space structure

The one property that does also hold for $L^2(\mathbb{R})$ is that we have an inner product

$$\langle f, g \rangle := \int_{-\infty}^{\infty} f(x) \overline{g(x)} dx \quad (4.1)$$

for $f, g \in L^2(\mathbb{R})$. The definition (4.1) is well-defined because, by Hölder's inequality,

$$|\langle f, g \rangle| = \left| \int_{-\infty}^{\infty} f(x) \overline{g(x)} dx \right| \leq \int_{-\infty}^{\infty} |f(x)g(x)| dx = \|f \cdot g\|_1 \leq \|f\|_2 \|g\|_2$$

which is finite since $f, g \in L^2(\mathbb{R})$. The norm induced by (4.1) is precisely the L^2 -norm, since $\|f\|_2^2 = \langle f, f \rangle$. Since $L^2(\mathbb{R})$ is complete with respect to this norm, it is a *Hilbert space*.

Definition 4.1. The L^2 -norm of a (measurable) function $f \rightarrow \mathbb{C}$ is called the *energy* of f . A *finite-energy signal* is a function in $L^2(\mathbb{R})$. ◀

4.1.2 Definition and properties

For $f \in L^2(\mathbb{R})$, the integral (3.1) may not converge.

Theorem 4.1. Let $f \in L^2(\mathbb{R})$. Then,

$$\hat{f}(\xi) := \lim_{R \rightarrow \infty} \int_{-R}^R f(x) e^{-2\pi i \xi x} dx \quad (4.2)$$

converges for almost every $\xi \in \mathbb{R}$ and is in $L^2(\mathbb{R})$. ◀

Theorem 4.2. The linear operator $\mathcal{F}: f \in L^2(\mathbb{R}) \mapsto \hat{f} \in L^2(\mathbb{R})$ satisfies:

- (i) $\mathcal{F}: L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ is bounded, injective and surjective. Its inverse is also bounded.
- (ii) For every $f, g \in L^2(\mathbb{R})$, $\langle f, \hat{g} \rangle = \langle f, g \rangle$, in particular $\|f\|_2 = \|\hat{f}\|_2$, i.e., $\mathcal{F}$ is a surjective isometry of $L^2(\mathbb{R})$.¹ ◀

Theorem 4.3 (Inverse). Let $f \in L^2(\mathbb{R})$. Then, $f(x) \stackrel{L^2}{=} \lim_{R \rightarrow \infty} \int_{-R}^R \hat{f}(\xi) e^{2\pi i \xi x} d\xi$. ◀

Different from $L^1(\mathbb{R})$, where $\hat{f}$ is continuous and defined for every $\xi \in \mathbb{R}$, the Fourier transform of $f \in L^2(\mathbb{R})$ does not have the same nice behavior. In general, the Fourier transform of $f \in L^2(\mathbb{R})$ is not a continuous function, and may even be infinite for some $\xi \in \mathbb{R}$.

Example 4.1. For $f(x) = \frac{\sin(\pi x)}{\pi x} \in L^2(\mathbb{R})$, we have

$$\hat{f}(\xi) = \begin{cases} 0 & |\xi| > 1/2 \\ 1/2 & |\xi| = 1/2 \\ 1 & |\xi| < 1/2 \end{cases}$$

which is not continuous. ◀

Example 4.2. Take $f(x) = \frac{1}{(1+|x|)^\alpha}$ for $1/2 < \alpha < 1$. Then, $f \in L^2(\mathbb{R}) \setminus L^1(\mathbb{R})$ and

$$\hat{f}(0) = \lim_{R \rightarrow \infty} \int_{-R}^R f(x) dx = \int_{-\infty}^{\infty} f(x) dx = \|f\|_1 = \infty$$

by the monotone convergence theorem (since $f \geq 0$). ◀

¹in fact, a unitary operator

4.2 Fourier transform on L^p , $1 < p < 2$

For $p = 2$, we have the complete characterization $\mathcal{F}(L^2(\mathbb{R})) = L^2(\mathbb{R})$. On the other hand, for $p = 1$, $\mathcal{F}(L^1(\mathbb{R}))$ is far from being $L^\infty(\mathbb{R})$. By Fact 3.1 and Theorem 3.6, we know that

$$\mathcal{F}(L^1(\mathbb{R})) \subseteq C_0(\mathbb{R}) \cap C_{\text{uni}}(\mathbb{R}) \subsetneq L^\infty(\mathbb{R})$$

where $C_{\text{uni}}(\mathbb{R})$ denotes the space of uniformly continuous functions on $\mathbb{R}$. The first inclusion is also strict, i.e.,

$$\mathcal{F}(L^1(\mathbb{R})) \subsetneq C_0(\mathbb{R}) \cap C_{\text{uni}}(\mathbb{R}) \subsetneq L^\infty(\mathbb{R}) \quad (4.3)$$

since there exist functions in $C_0(\mathbb{R}) \cap C_{\text{uni}}(\mathbb{R})$ which are not the Fourier transform of any L^1 -function. Unlike $p = 2$, the image of $L^1(\mathbb{R})$ under the Fourier transform does not admit a simple characterization.

In general, the behavior of $\mathcal{F}$ on $L^p(\mathbb{R})$ is subtle. It can be proven that if $f \in L^p(\mathbb{R})$, $1 < p < 2$, there exist (non-unique) $f_1 \in L^1(\mathbb{R})$ and $f_2 \in L^2(\mathbb{R})$ such that $f = f_1 + f_2$. Hence, we may define the Fourier transform of f as $\hat{f} = \hat{f}_1 + \hat{f}_2$. Moreover, this definition does not depend on the choice of f_1 and f_2 . We have the following celebrated result regarding boundedness:

Theorem 4.4 (Hausdorff-Young). The Fourier transform $\mathcal{F}$ maps $L^p(\mathbb{R})$ to $L^{p'}(\mathbb{R})$ for every $1 \leq p \leq 2$, where $p' = \frac{p}{p-1}$ is the conjugate exponent of p . Moreover,

$$\|\hat{f}\|_{p'} \leq \left(\frac{p^{1/p}}{(p')^{1/p'}} \right)^{1/2} \|f\|_p \quad (4.4)$$

for every $f \in L^p(\mathbb{R})$. Also known as the **Babenko–Beckner inequality**. $\triangleleft$

The mapping $\mathcal{F}: L^p(\mathbb{R}) \rightarrow L^{p'}(\mathbb{R})$, $1 \leq p \leq 2$, is surjective if and only if $p = 2$. In fact, $\mathcal{F}(L^p(\mathbb{R}))$ admits an explicit characterization only in this case. Moreover, for every $p > 2$ there does always exist a function $f \in L^p(\mathbb{R})$ such that $\hat{f} \notin L^q(\mathbb{R})$ for any $0 < q \leq \infty$. Even worse, one can find $f \in L^p(\mathbb{R})$ such that $\hat{f} \notin L^1_{\text{loc}}(\mathbb{R})$, where $L^1_{\text{loc}}(\mathbb{R})$ denotes the space of measurable functions $f: \mathbb{R} \rightarrow \mathbb{C}$ satisfying

$$\int_K |f(x)| dx < \infty$$

for every compact $K \subseteq \mathbb{R}$ (locally integrable functions). In order to develop a Fourier theory for L^p functions beyond the case $p = 2$, one is naturally led to the framework of *distributions*.

4.3 Multivariate Fourier transform

For a multivariate function $f: \mathbb{R}^d \rightarrow \mathbb{C}$, $f \in L^1(\mathbb{R}^d)$, we define

$$\hat{f}(\boldsymbol{\xi}) := \int_{\mathbb{R}^d} f(\mathbf{x}) e^{-2\pi i \boldsymbol{\xi} \cdot \mathbf{x}} d\mathbf{x} \quad (4.5)$$

where $\boldsymbol{\xi} \cdot \mathbf{x} = \sum_{j=1}^d \xi_j x_j$ is the standard inner product in $\mathbb{R}^d$.

The lower dilation must be re-defined as

$$f_\lambda(\mathbf{x}) = \frac{1}{\lambda^d} f\left(\frac{\mathbf{x}}{\lambda}\right)$$

and Gaussians are replaced by multivariate Gaussians $\varphi(\mathbf{x}) = e^{-\pi \|\mathbf{x}\|^2}$.

If $f \in C^k(\mathbb{R}^d)$ and $|\boldsymbol{\alpha}| = \alpha_1 + \dots + \alpha_d \leq k$, we define

$$D^\alpha f(\mathbf{x}) = \frac{\partial^{|\boldsymbol{\alpha}|}}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}} f(\mathbf{x})$$

where $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}^d$ is a multi-index. Under this notation, Equation 3.20 becomes

$$\widehat{D^\alpha f}(\boldsymbol{\xi}) = (2\pi i \boldsymbol{\xi})^\alpha \hat{f}(\boldsymbol{\xi})$$

where $(2\pi i \boldsymbol{\xi})^\alpha = (2\pi i)^{|\boldsymbol{\alpha}|} \xi_1^{\alpha_1} \dots \xi_d^{\alpha_d}$.

5 PDEs

5.1 Sobolev spaces

Consider the ODE

$$\boxed{-u_{xx} + u = f} \quad (5.1)$$

for a fixed forcing $f \in L^1(\mathbb{R})$. Assuming that $u \in L^1(\mathbb{R})$, we take the Fourier transform to obtain

$$(4\pi^2\xi^2 + 1)\hat{u}(\xi) \stackrel{(3.20)}{=} \hat{f}(\xi) \implies \hat{u}(\xi) = \frac{1}{1 + 4\pi^2\xi^2} \cdot \hat{f}(\xi)$$

whence,

$$u = \mathcal{F}^{-1} \left(\frac{1}{1 + 4\pi^2(\cdot)^2} \cdot \hat{f} \right) \stackrel{(3.16)}{=} \mathcal{F}^{-1} \left(\frac{1}{1 + 4\pi^2(\cdot)^2} \right) * f$$

By Example 3.4

$$\frac{1}{1 + \pi^2(2\xi)^2} = \mathcal{F}(e^{-2|\cdot|})(2\xi) = \frac{1}{2}\mathcal{F}(e^{-|\cdot|})(\xi)$$

where we used Equation 3.12. Therefore, $\mathcal{F}^{-1} \left(\frac{1}{1 + 4\pi^2(\cdot)^2} \right) = \frac{1}{2}e^{-|\cdot|}$.

Thus, the solution of Equation 5.1 can be expressed as

$$u(x) = \frac{1}{2} \int_{-\infty}^{\infty} e^{-|x-y|} f(y) dy$$

Definition 5.1 (Lebesgue-Sobolev space). For $s \in \mathbb{R}$, we define

$$\boxed{H^s(\mathbb{R}) := \{f \in L^2(\mathbb{R}) : \Lambda_s f \in L^2(\mathbb{R})\}} \quad (5.2)$$

where

$$\Lambda_s f = \mathcal{F}^{-1}((1 + 4\pi^2(\cdot)^2)^{s/2} \cdot \hat{f}) \quad (5.3)$$

denotes the Bessel-Sobolev potential of order s . ◀

5.1.1 Norm

A norm is naturally defined on $H^s(\mathbb{R})$ by $\|f\|_{H^s} := \|\Lambda_s f\|_2$. Recall that the Fourier and its inverse are isometries of $L^2(\mathbb{R})$, so

$$\|f\|_{H^s} = \|\Lambda_s f\|_2 = \|\mathcal{F}^{-1}((1 + 4\pi^2(\cdot)^2)^{s/2} \cdot \hat{f})\|_2 = \|(1 + 4\pi^2(\cdot)^2)^{s/2} \cdot \hat{f}\|_2$$

whence

$$\boxed{\|f\|_{H^s} = \left(\int_{-\infty}^{\infty} (1 + 4\pi^2\xi^2)^s |\hat{f}(\xi)|^2 d\xi \right)^{1/2}} \quad (5.4)$$

Trivially, $\Lambda_0 f = f$ for every $f \in L^2(\mathbb{R})$, so $H^0(\mathbb{R}) = L^2(\mathbb{R})$. Moreover, if $s = k \in \mathbb{N}_{\geq 1}$,

$$\begin{aligned} \|f\|_{H^k}^2 &= \int_{-\infty}^{\infty} (1 + 4\pi^2\xi^2)^k |\hat{f}(\xi)|^2 d\xi = \int_{-\infty}^{\infty} \left(\sum_{j=0}^k \binom{k}{j} (4\pi^2\xi^2)^j \right) |\hat{f}(\xi)|^2 d\xi \\ &= \sum_{j=0}^k \binom{k}{j} \int_{-\infty}^{\infty} (4\pi^2\xi^2)^j |\hat{f}(\xi)|^2 d\xi = \|f\|_2^2 + \sum_{j=1}^k \binom{k}{j} \int_{-\infty}^{\infty} (4\pi^2\xi^2)^j |\hat{f}(\xi)|^2 d\xi \\ &= \|f\|_2^2 + \sum_{j=1}^k \binom{k}{j} \int_{-\infty}^{\infty} |(2\pi i\xi)^j \hat{f}(\xi)|^2 d\xi \\ &\stackrel{(3.20)}{=} \|f\|_2^2 + \sum_{j=1}^k \binom{k}{j} \int_{-\infty}^{\infty} |\widehat{f^{(j)}}(\xi)|^2 d\xi \\ &= \|f\|_2^2 + \sum_{j=1}^k \binom{k}{j} \|\mathcal{F}(f^{(j)})\|_2^2 \\ &= \|f\|_2^2 + \sum_{j=1}^k \binom{k}{j} \|f^{(j)}\|_2^2 \end{aligned}$$

Therefore,

$$\|f\|_{H^k} = \sqrt{\|f\|_2^2 + \sum_{j=1}^k \binom{k}{j} \|f^{(j)}\|_2^2} \quad (5.5)$$

We have the estimate

$$\sqrt{\|f\|_2^2 + \sum_{j=1}^k \|f^{(j)}\|_2^2} \leq \|f\|_{H^k} \leq \binom{k}{\lfloor k/2 \rfloor}^{\frac{1}{2}} \sqrt{\|f\|_2^2 + \sum_{j=1}^k \|f^{(j)}\|_2^2} \quad (5.6)$$

since $1 \leq \binom{k}{j} \leq \binom{k}{\lfloor k/2 \rfloor}$ for every $0 \leq j \leq k$.

Furthermore, for $x_1, \dots, x_k \geq 0$ and $k \in \mathbb{N}_{\geq 1}$,

$$\frac{1}{k} (x_1 + \dots + x_k)^2 \leq x_1^2 + \dots + x_k^2 \leq (x_1 + \dots + x_k)^2 \quad (5.7)$$

as can be proven by induction on k .

Combining (5.7) and (5.6), we obtain

$$\frac{1}{\sqrt{1+k}} \sum_{j=0}^k \|f^{(j)}\|_2 \leq \|f\|_{H^k} \leq \binom{k}{\lfloor k/2 \rfloor}^{\frac{1}{2}} \sum_{j=0}^k \|f^{(j)}\|_2 \quad (5.8)$$

where we denoted $f^{(0)} = f$ for compactness.

5.1.2 Weak derivative

Definition 5.2 clarifies the notion of derivative appearing in the next sections. Recall that $L_{\text{loc}}^1(\mathbb{R})$ is the space of measurable functions $f : \mathbb{R} \rightarrow \mathbb{C}$ such that

$$\int_K |f(x)| dx < \infty$$

for every compact K in $\mathbb{R}$.

Definition 5.2 (weak derivative). Let $f \in L^2(\mathbb{R})$. We say that f has weak derivative of order k if there exists a function $F \in L_{\text{loc}}^1(\mathbb{R})$ such that for every $g \in C_c^\infty(\mathbb{R})$,

$$\int_{-\infty}^{\infty} F(x)g(x)dx = (-1)^k \int_{-\infty}^{\infty} f(x)g^{(k)}(x)dx \quad (5.9)$$

The function F is called the k -th weak derivative of f and will be denoted by $f^{(k)}$. ◀

Theorem 5.1. Let $k \in \mathbb{N}$. Then, $f \in H^k(\mathbb{R})$ if and only if f has weak derivatives up to order k and for every $j \leq k$, $f^{(j)} \in L^2(\mathbb{R})$. ◀

5.2 Heat equation

Consider the Cauchy problem

$$\begin{cases} \partial_t u(t, x) = \partial_{xx}^2 u(t, x) & t > 0, x \in \mathbb{R} \\ u(0, x) = u_0(x) & x \in \mathbb{R} \end{cases} \quad (5.10)$$

where the initial datum u_0 is assumed to be sufficiently regular.

To solve (5.10), take FT wrt the space variable x . It commutes with ∂_t and to the RHS we apply Theorem 3.5:

$$\begin{cases} \partial_t \hat{u}(t, \xi) = -4\pi^2 \xi^2 \hat{u}(t, \xi) & t > 0, \xi \in \mathbb{R} \\ \hat{u}(0, \xi) = \widehat{u_0}(\xi) & \xi \in \mathbb{R} \end{cases} \quad (5.11)$$

For every fixed $\xi \in \mathbb{R}$, (5.11) is a first-order ODE in t with general solution given by

$$\hat{u}(t, \xi) = C e^{-4\pi^2 t \xi^2} \quad (5.12)$$

where the constant C is chosen such that the initial condition is satisfied:

$$[\hat{u}(t, \xi)]_{t=0} \stackrel{!}{=} \widehat{u_0}(\xi) \implies C = \widehat{u_0}(\xi)$$

whence the solution of (5.11) is

$$\hat{u}(t, \xi) = e^{-4\pi^2 t \xi^2} \widehat{u_0}(\xi) \quad (5.13)$$

By taking the inverse Fourier transform, we obtain

$$\begin{aligned} u(t, x) &= \mathcal{F}^{-1} \left(e^{-4\pi^2 t (\cdot)^2} \widehat{u_0} \right) (x) \\ &\stackrel{(3.16)}{=} \left(\mathcal{F}^{-1} \left(e^{-4\pi^2 t (\cdot)^2} \right) * \mathcal{F}^{-1}(\widehat{u_0}) \right) (x) \\ &\stackrel{(3.13)}{=} \left(\frac{1}{\sqrt{4\pi t}} e^{-\frac{(\cdot)^2}{4t}} * u_0 \right) (x) \end{aligned} \quad (5.14)$$

where we used Corollary 3.12, (3.29), and Equation 3.12 with $\lambda = \sqrt{4\pi t}$

In conclusion,

$$u(t, x) = (K_t * u_0)(x) \quad (5.15)$$

where $K_t(x) = \frac{1}{\sqrt{4\pi t}} e^{-\frac{x^2}{4t}}$, $t > 0$, $x \in \mathbb{R}$, is called **heat kernel**.

5.3 Wave equation

Consider the Cauchy problem

$$\begin{cases} \partial_{tt}^2 u(t, x) = \partial_{xx}^2 u(t, x) & t > 0, x \in \mathbb{R} \\ u(0, x) = u_0(x) & x \in \mathbb{R} \\ \partial_t u(0, x) = v_0(x) & x \in \mathbb{R} \end{cases} \quad (5.16)$$

where the initial data u_0, v_0 are assumed to be sufficiently regular.

To solve (5.16), take FT wrt the space variable x . It commutes with ∂_t and to the RHS we apply Theorem 3.5:

$$\begin{cases} \partial_{tt}^2 \hat{u}(t, \xi) = -4\pi^2 \xi^2 \hat{u}(t, \xi) & t > 0, \xi \in \mathbb{R} \\ \hat{u}(0, \xi) = \widehat{u_0}(\xi) & \xi \in \mathbb{R} \\ \partial_t \hat{u}(0, \xi) = \widehat{v_0}(\xi) & \xi \in \mathbb{R} \end{cases} \quad (5.17)$$

For every fixed $\xi \in \mathbb{R}$, (5.17) is a second-order ODE in t . Searching for solutions of the form $\hat{u}(t, \xi) = e^{\lambda t}$ yields

$$\lambda^2 + 4\pi^2 \xi^2 = 0$$

hence $\lambda = \pm 2\pi i |\xi|$, and thus, for $\xi \neq 0$ the general solution² is given by

$$\hat{u}(t, \xi) = A(\xi) e^{-2\pi i |\xi| t} + B(\xi) e^{2\pi i |\xi| t} \quad (5.18)$$

where $A(\xi)$ and $B(\xi)$ are obtained by imposing the initial conditions in (5.17):

$$A(\xi) = \frac{1}{2} \widehat{u_0}(\xi) - \frac{1}{2} \frac{\widehat{v_0}(\xi)}{2\pi i |\xi|} \quad B(\xi) = \frac{1}{2} \widehat{u_0}(\xi) + \frac{1}{2} \frac{\widehat{v_0}(\xi)}{2\pi i |\xi|}$$

By associativity and the identities $\frac{e^{i\theta} + e^{-i\theta}}{2} = \cos(\theta)$ and $\frac{e^{i\theta} - e^{-i\theta}}{2i} = \sin(\theta)$, we obtain

$$\hat{u}(t, \xi) = \cos(2\pi |\xi| t) \widehat{u_0}(\xi) + \frac{\sin(2\pi |\xi| t)}{2\pi |\xi|} \widehat{v_0}(\xi) \quad (5.19)$$

with the inverse

$$u(t, x) = \mathcal{F}^{-1}(\cos(2\pi t |\cdot|) \widehat{u_0}) (x) + \mathcal{F}^{-1} \left(\frac{\sin(2\pi t |\cdot|)}{2\pi |\cdot|} \widehat{v_0} \right) (x) \quad (5.20)$$

For the first term, using $\cos(\theta) = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$ and the inversion formula:

$$\mathcal{F}^{-1}(\cos(2\pi t |\cdot|) \widehat{u_0}) (x) = \frac{u_0(x+t) + u_0(x-t)}{2} \quad (5.21)$$

²For $\xi = 0$ the solution is different (since the characteristic polynomial has a double root), but for our purposes we can ignore that, since the (inverse) FT does not care about a set of measure zero.

For the second term, we use (3.4), (3.16) and (3.28):

$$\begin{aligned}
 \mathcal{F}^{-1}\left(\frac{\sin(2\pi t|\cdot|)}{2\pi|\cdot|}\widehat{v}_0\right)(x) &= \frac{1}{2}(\chi_{[-t,t]} * v_0)(x) \\
 &= \frac{1}{2}\int_{-\infty}^{\infty}\chi_{[-t,t]}(x-y)v_0(y)\,dy \\
 &= \frac{1}{2}\int_{x-t}^{x+t}v_0(y)\,dy
 \end{aligned} \tag{5.22}$$

In conclusion, combining (5.20), (5.21), and (5.22), we obtain **d'Alembert's formula**

$$u(t, x) = \frac{u_0(x+t) + u_0(x-t)}{2} + \frac{1}{2}\int_{x-t}^{x+t}v_0(y)\,dy \tag{5.23}$$

for $t > 0$ and $x \in \mathbb{R}$.